USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

[22CSBS228]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

OBJECT ORIENTED PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART- A			**05 X 02=10**
1.	**a.**	Differentiate procedural oriented and object oriented programming	L3	CO1	**2**
	b.	the syntax for switch statement in C++	L1	CO2	**2**
	c.	Define polymorphism	L1	CO3	**2**
	d.	List out the advantages of generic programming.	L1	CO4	**2**
	e.	What is meant by unified modelling language?	L1	CO5	**2**
		PART-B			**06 X 05=30**
2.	**a.**	Define pointer. Write a C++ program to sort an array in ascending order using pointer.	L3	CO1	**05**
		OR			
	b.	Explain the following with an example C++ program **i.** Call by Value **ii.** Call by Reference	L3	CO1	**05**
3.	**a.**	Define class. Explain with an example how the member functions of the class can be defined?	L3	CO2	**05**
		OR			
	b.	Write a C++ program to illustrate the types of constructor.	L3	CO2	**05**
4.	**a.**	Define Operator Overloading? Explain it with an example C++ program.	L3	CO3	**05**
		OR			
	b.	Explain single inheritance with an example C++ program.	L3	CO3	**05**
5.	**a.**	Explain the function template with the C++ program	L3	CO4	**05**
		OR			
	b.	Explain the hierarchy of the C++ stream classes with the neat illustrations	L3	CO4	**05**
6.	**a.**	What is meant by class relationships? Explain	L3	CO5	**05**
		OR			
	b.	Define Activity diagram? Draw the activity diagram for ATM Process.	L3	CO5	**05**

7.	a.	Define Strings. Explain string handling functions with an example.	L3	CO1	05
		OR			
	b.	Explain the different visibility modes for the class members in C++.	L2	CO2	05
		PART-C			**06 X 10=60**
8.	a.	Explain the various types of numeric constants in C++ with the neat illustrations.	L2	CO1	10
		OR			
	b.	What are control structures? Explain iterative control structures with an example program.	L2	CO1	10
9.	a.	List the properties of friend function and explain it with an example	L2	CO2	10
		OR			
	b.	Explain constructor overloading with an example	L2	CO2	10
10.	a.	State the advantage of using virtual function. Explain Virtual Function with an example C++ program.	L3	CO3	10
		OR			
	b.	Describe error handling mechanism in C++?	L3	CO3	10
11.	a.	Write a C++ program to explain any two file handling functions?	L3	CO4	10
		OR			
	b.	Define Dynamic Binding. Write a C++ program to explain polymorphism through dynamic binding.	L3	CO4	10
12.	a.	Explain the UML models in object oriented modeling and design?	L3	CO5	10
		OR			
	b.	**i.** Differentiate function oriented design and object oriented design? **ii.** Design a use case diagram for online shopping system	L3	CO5	10
13.	a.	Explain the derived data types in C++ with the examples	L3	CO1	10
		OR			
	b.	Explain function overloading with an example C++ program	L3	CO4	10